Samuel Besley

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PROFESSIONAL SUMMARY

Data scientist and research software developer with a B.S. in Data Science from Emory University. Builds tools that put rigorous analysis in researchers' hands: an offline study-data workspace with statistics validated against R, NLP pipelines using LLM-based extraction, classical ML models, and a Rust/TypeScript data-visualization studio. Seeking Data Scientist and ML Engineer roles where applied ML has direct product or research impact.

EDUCATION

Emory University Atlanta, GA
B.S. in Data Science, International Studies Track May 2026
Relevant Coursework: Machine Learning (QTM 347), Natural Language Processing (CS/QTM/LING 329), Data Structures, Applied Regression, East European Politics, International Relations & Conflict

EXPERIENCE

Research Software Developer Oct 2026 – Present
BARI-CNS Lab, Emory University Atlanta, GA

- Built the lab's offline research workspace, an Electron/React/TypeScript desktop app for macOS and Windows that imports longitudinal study spreadsheets into SQLite while preserving source attribution, QC highlights, and comments, so researchers can explore data without SQL
- Implemented on-demand statistics (Welch t, Mann–Whitney, ANOVA, Kruskal–Wallis with effect sizes and Holm-adjusted comparisons; Pearson/Spearman/Kendall; regression bands; correlation matrix; PCA), validated against R 4.3, with every step recorded as a reproducible R-syntax analysis log
- Added a read-only MCP server that lets AI assistants query aggregate results with small-cell suppression and access logging; shipped four releases in five days, backed by 141 unit/integration tests and Playwright E2E tests

Photographer & Communications Contributor Aug 2023 – May 2024
Oxford College of Emory University Oxford, GA

- Conducted interviews and produced photo content for Emory's official communications team, published across institutional social media and web channels

Server & Bartender Aug 2022 – Aug 2024
Seven Acre Dairy; Everly Madison, WI

PROJECTS

Albi: Data-Visualization Studio | Rust, TypeScript, Svelte Fall 2026

- Built a macOS app for editorial data notebooks that export as one self-contained, offline HTML file; a Rust core runs the data engine, validation, and export, and one TypeScript renderer draws charts in both app and exports
- Engineered a data layer for CSV, JSON, Parquet, spreadsheets, SQLite, and PostgreSQL with 13 recorded preparation steps; a 1M-row CSV imports in ~1.3 s without freezing the UI
- Shipped a 38-tool MCP server so AI assistants edit through the same validated, undoable commands as the UI; exports pass automated WCAG 2.2 AA audits

EpsTracked: Document Analysis Dashboard | Python, React, TypeScript Spring 2026

- Built and deployed a full-stack NLP app (eps-tracked.vercel.app) over the public Epstein document corpus, with a React/TypeScript multi-view dashboard; CS/QTM/LING 329 (Computational Linguistics)
- Engineered a Python pipeline using LLM-based structured extraction (Anthropic API) that pulled 2,599 events from 787 document threads, classified them into six event types with likelihood scores, and resolved 515 unique entities

NYC Airbnb Price Prediction | Python, scikit-learn, pandas Spring 2026

- Trained a price model on 48,000+ NYC listings combining structured features with 25 engineered NLP features from listing names; benchmarked OLS, Ridge, LASSO, and Random Forest
- Achieved R^2 of 0.53 and MAE of ~$60 on log-transformed prices; showed text features add signal beyond structured data

Besley's Lab: ML From Scratch | Next.js, TypeScript 2026 – Present

- Implemented nine interactive ML demos with no ML libraries (gradient descent, k-means, SVM, Q-learning, more), each tested

SKILLS

ML / NLP: Regression, Random Forest, Ridge/LASSO, PCA, Transformers (BERT), RNNs, LLM-based extraction, feature engineering
Statistics: Hypothesis tests (t, Mann–Whitney, ANOVA, Kruskal–Wallis), effect sizes, multiple-comparison correction
Programming: Python (pandas, scikit-learn, NumPy, HuggingFace), TypeScript/JavaScript, SQL, R, Rust, Java
Tools: React, Next.js, Electron, SQLite, PostgreSQL, Vitest, Playwright, Git/GitHub, Jupyter, MCP
Languages: English (native), German (proficient), Spanish (proficient)